

**Presentation
on**

Spam Email Detection

Using Naive Bayes and Random Forest
Classifier

CSE 411: DATA SCIENCE

Submitted By

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Session: 2021-2022

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1. Introduction & Problem Statement

Overview

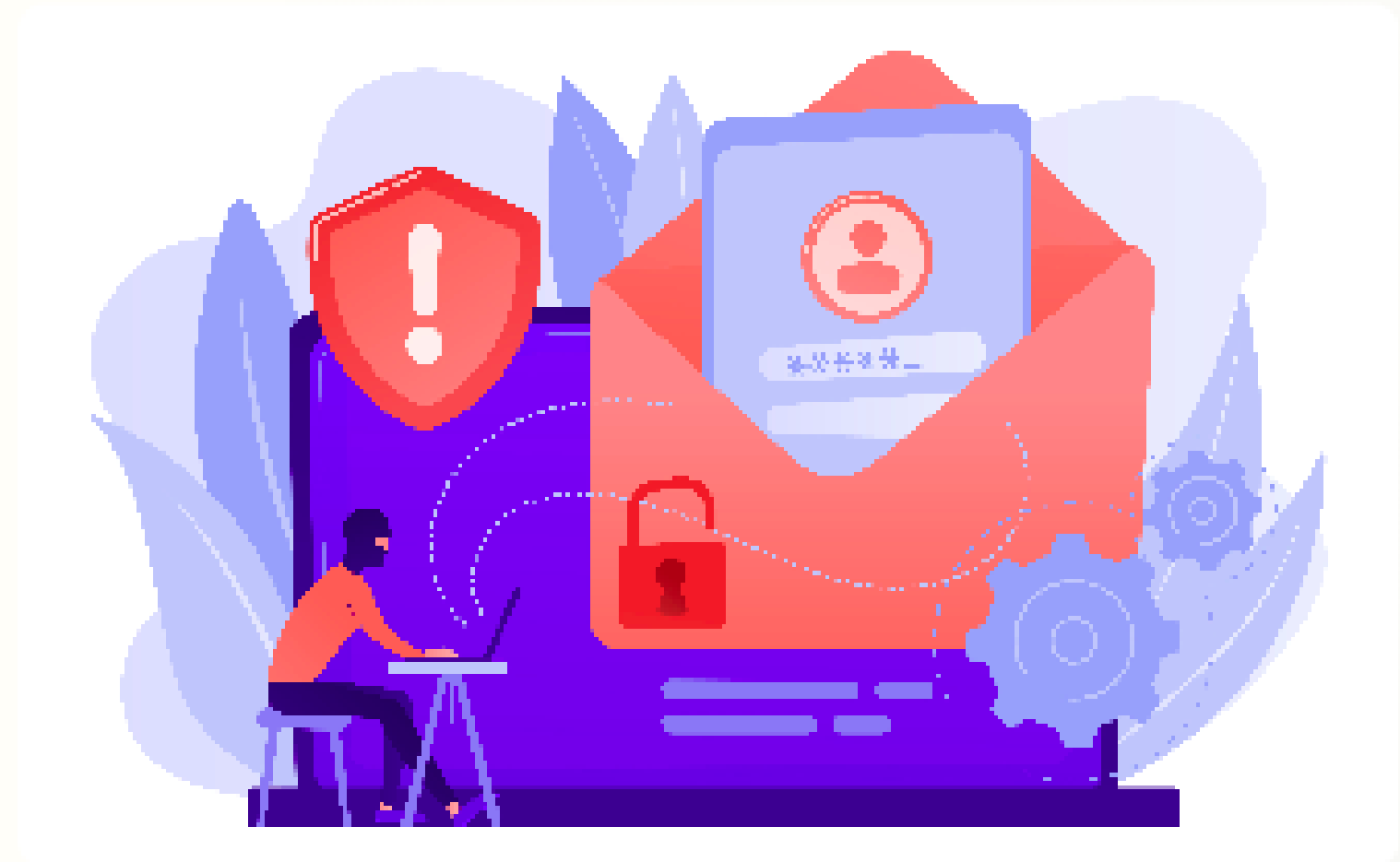
Email represents a cornerstone of digital communication. However, a massive volume of unwanted messages (Spam) containing fraud, phishing, and fake offers requires automated filtering to ensure security and efficiency.

The Problem

Manual identification is inefficient. An automated system is required to classify emails. This project implements two powerful Machine Learning models to solve this classification task effectively.

2. Objectives of the Assignment

- ✓ Load and analyze the *emails.csv* dataset.
- ✓ Apply Exploratory Data Analysis (EDA).
- ✓ Convert text data into numerical features using TF-IDF.
- ✓ Train and evaluate Naive Bayes and Random Forest.
- ✓ Conduct a performance comparative analysis.



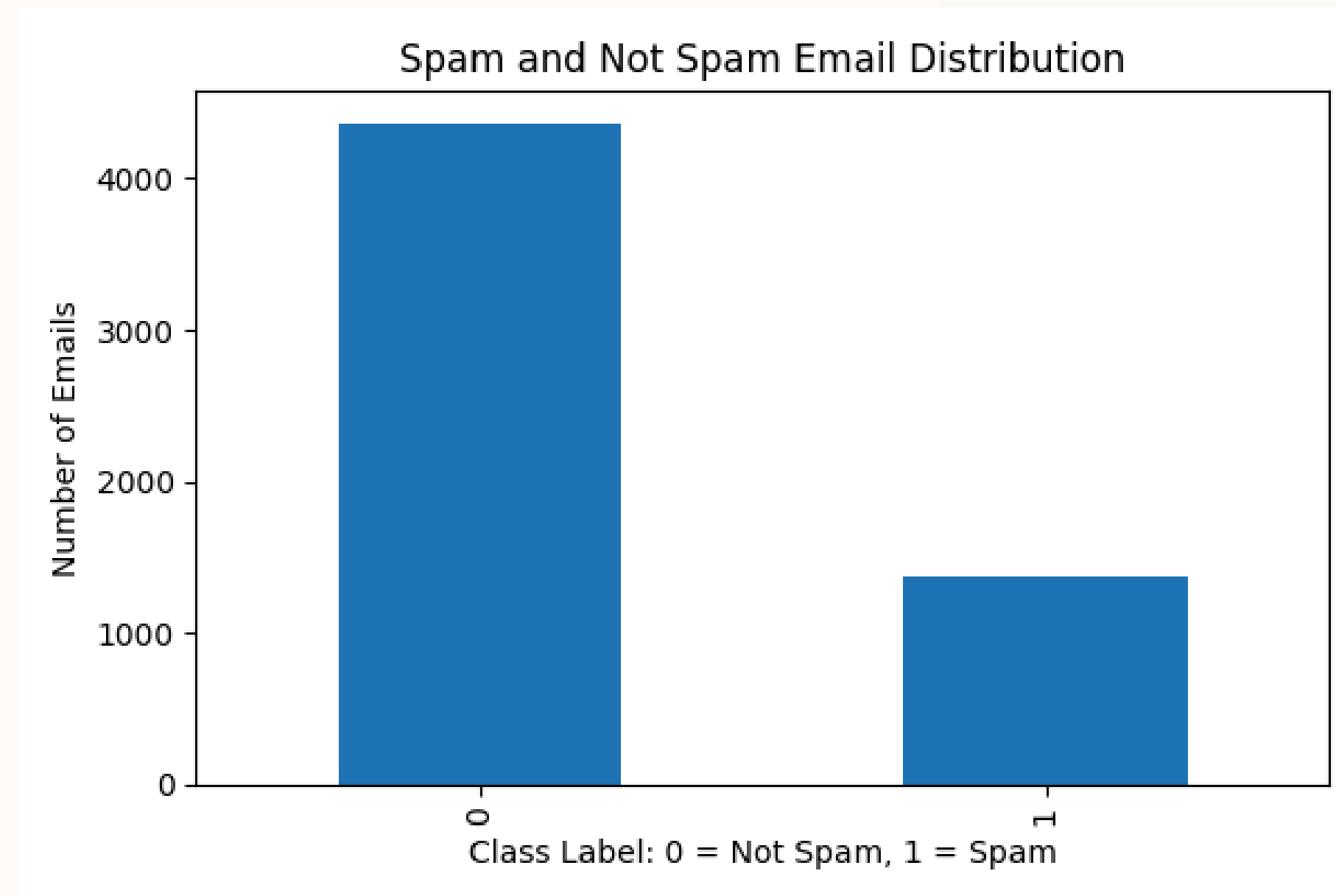
3. Dataset Summary

Property	Details
File Name	emails.csv
Total Records	5,728
Missing Values	0
Avg. Email Length	1,556 characters
Target Classes	2 (0: Ham, 1: Spam)



The dataset is slightly imbalanced, which is common for spam detection tasks, providing a realistic training scenario.

4. Exploratory Data Analysis



Distribution visualization confirms that ~24% of the dataset consists of spam messages.

5. Exploratory Data Analysis(cont.)

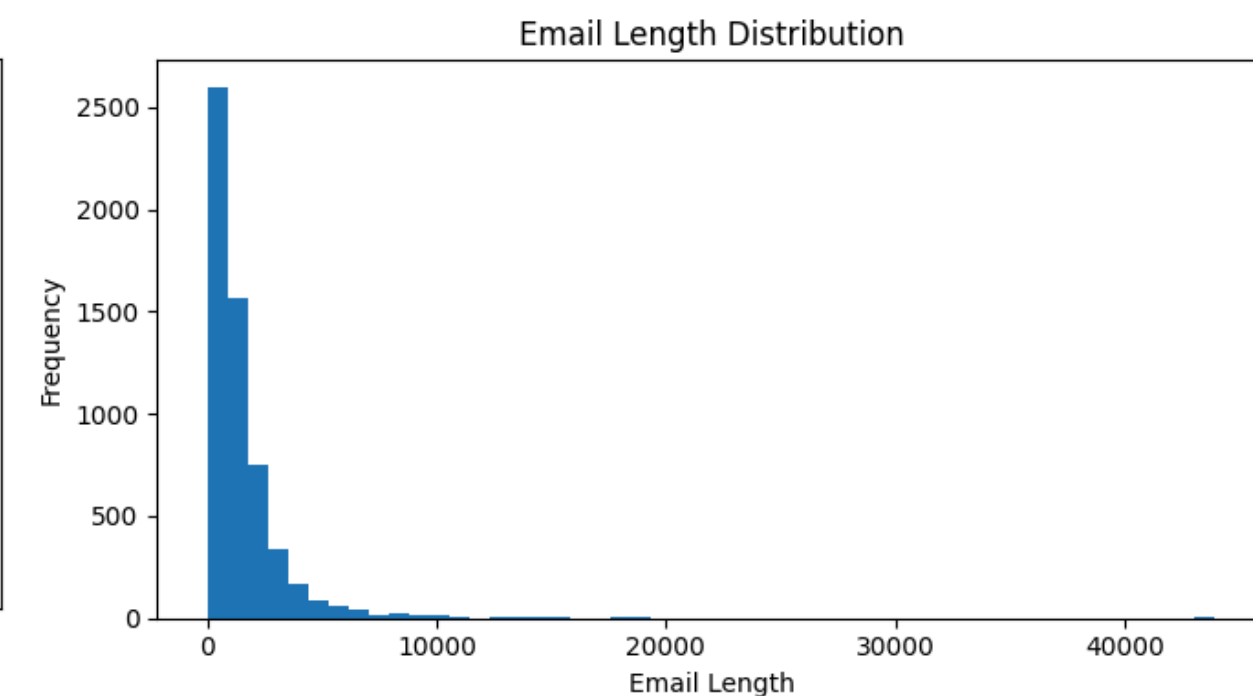
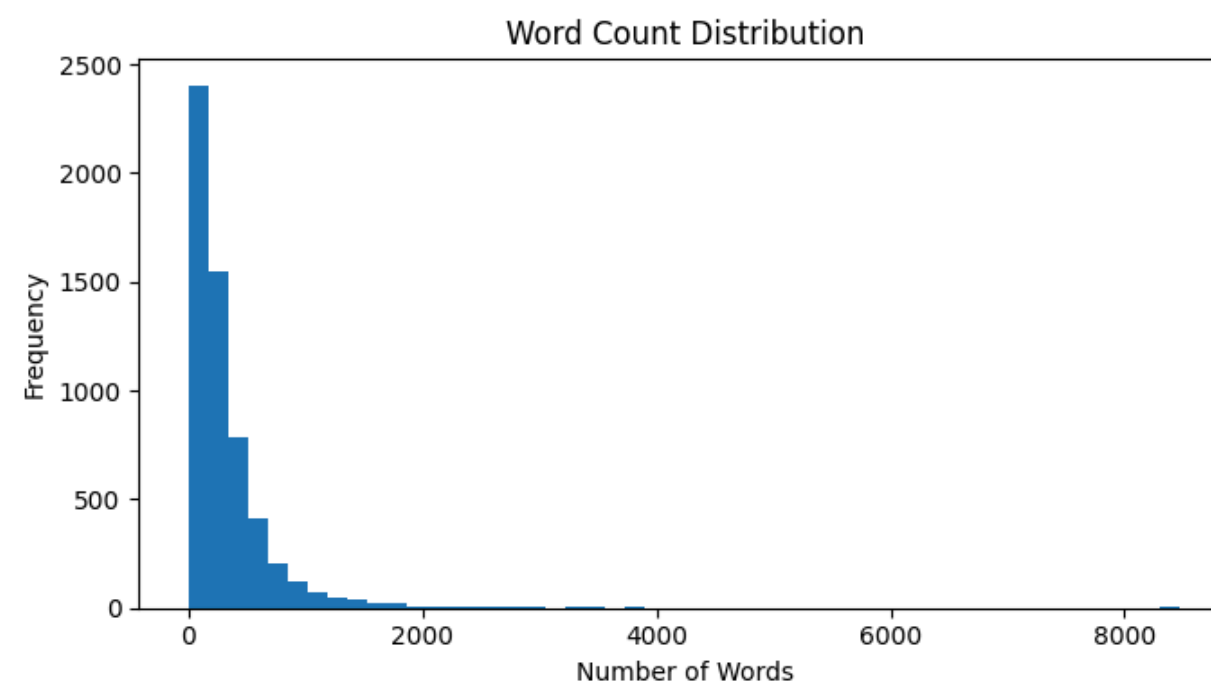
Email Length and Word Count Statistics:

```
count    5728.000000
mean     1556.768680
std      2042.649812
min       13.000000
25%      508.750000
50%      979.000000
75%     1894.250000
max     43952.000000
Name: email_length, dtype: float64
```

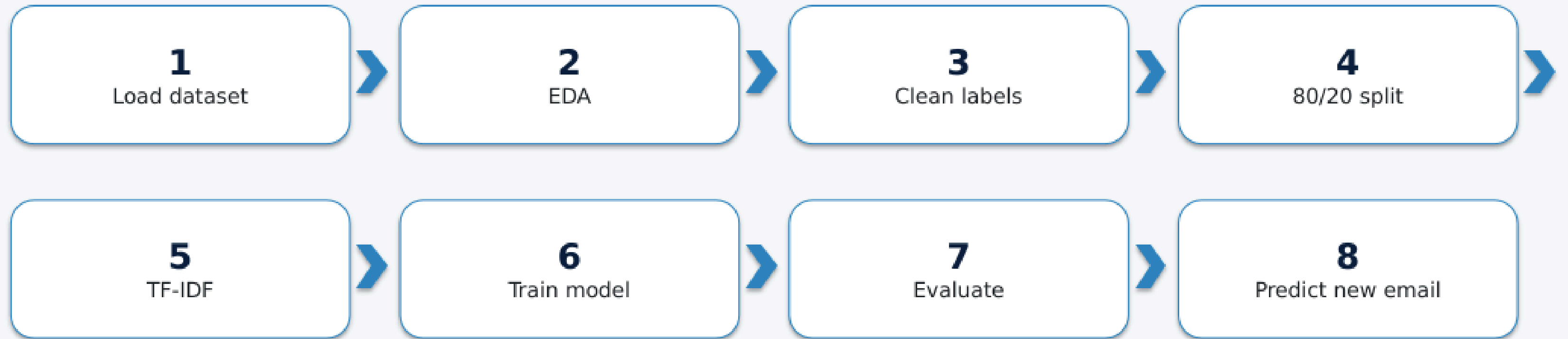
EDA observations

- The dataset is imbalanced, with more non-spam emails than spam emails.
- Email length and word count vary a lot across samples.
- These features suggest the model needs robust text representation.

```
Word Count Statistics:
count    5728.000000
mean     326.848638
std      418.777435
min        2.000000
25%      101.000000
50%      210.000000
75%      401.000000
max     8477.000000
Name: word_count, dtype: float64
```



6. System Workflow



Core idea

TF-IDF converts raw text into numerical vectors.
Naive Bayes and Random Forest are then trained on the same split for a fair comparison.

7. TF-IDF Vectorization

- **TF-IDF** stands for Term Frequency-Inverse Document Frequency
- It weighs words based on their uniqueness in a document relative to the corpus.

$$\text{TF-IDF} (t, d) = \text{TF} (t, d) \times \log \frac{N}{\text{DF} (t)}$$

Importance

This method ensures that common words like "the" or "is" are penalized, while specific spam keywords like "offer" or "win" gain higher importance.

8. Naive Bayes Classifier

Multinomial Naive Bayes

applied to sparse feature vector classification: c1 or c2

1. Bayes' theorem

$$p(c|x) = \frac{p(x|c)P(c)}{p(x)}$$

posterior ← likelihood
evidence ← prior

$$P(c) \sim \frac{\text{counts}}{N}$$

2. Predictor template

$$d = \underset{c}{\operatorname{argmax}} \left\{ \begin{array}{l} p(y|d) \sim P(c|y) \\ p(y_2|y) \sim p_2 \\ p(y_2|c) \sim p_2 \end{array} \right\} \rightarrow y^{\wedge}$$

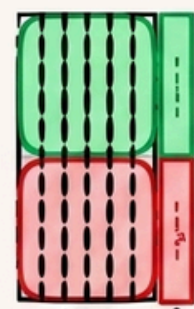
3 and 4. (Naive) multinomial hypothesis for simpler likelihood

$$p(w|x, y) \xrightarrow[\text{represents to evidence } \times]{p(w|y) \sim \text{Multinomial}(p(w|t, \theta_w, c, \theta_w)} p(w|y) = p(w_1|y) \dots p(w_j|y) = \text{count}(j)/N_c$$

return, lgn

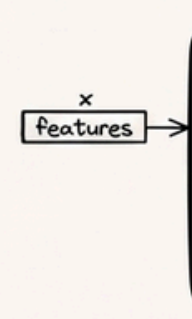
pulling to estimate ✓

5. Estimate multinomial parameters using train set


$$\hat{\theta}_1 = \frac{\sum \text{counts}_{c1}}{\sum \sum \text{counts}_{c1, c2}}$$
$$\hat{\theta}_2 = \frac{\sum \text{counts}_{c2}}{\sum \sum \text{counts}_{c1, c2}}$$

$\hat{\theta}_1 = \underset{c}{\operatorname{argmax}} \sum_j \text{counts}_{c1, c2}$
 $\hat{\theta}_2 = \underset{c}{\operatorname{argmax}} \sum_j \text{counts}_{c1, c2}$

6. Predict the class of a new sample


$$p(c|x) = P(w_1|c) \dots P(w_j|c) = p_1$$
$$p(c|x) = P(w_1|c_2) \dots P(w_j|c_2) = p_2$$

$\operatorname{argmax} \rightarrow y^{\wedge}$

A probabilistic model based on Bayes' Theorem. It assumes feature independence to simplify calculations.

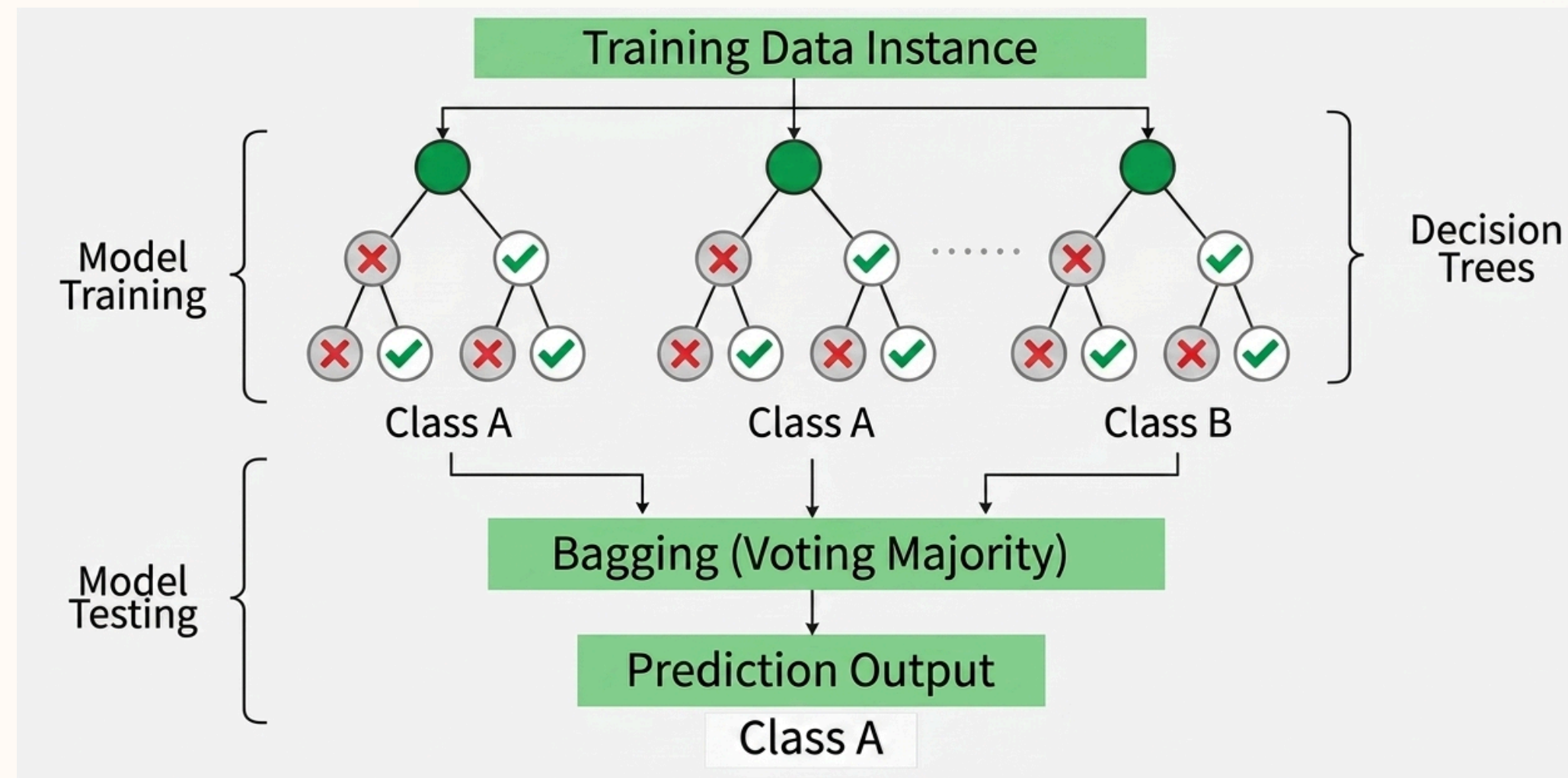
$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

- ✓ Extremely fast training speed.
- ✓ High performance on text data.
- ✓ **Accuracy: 99.04%**

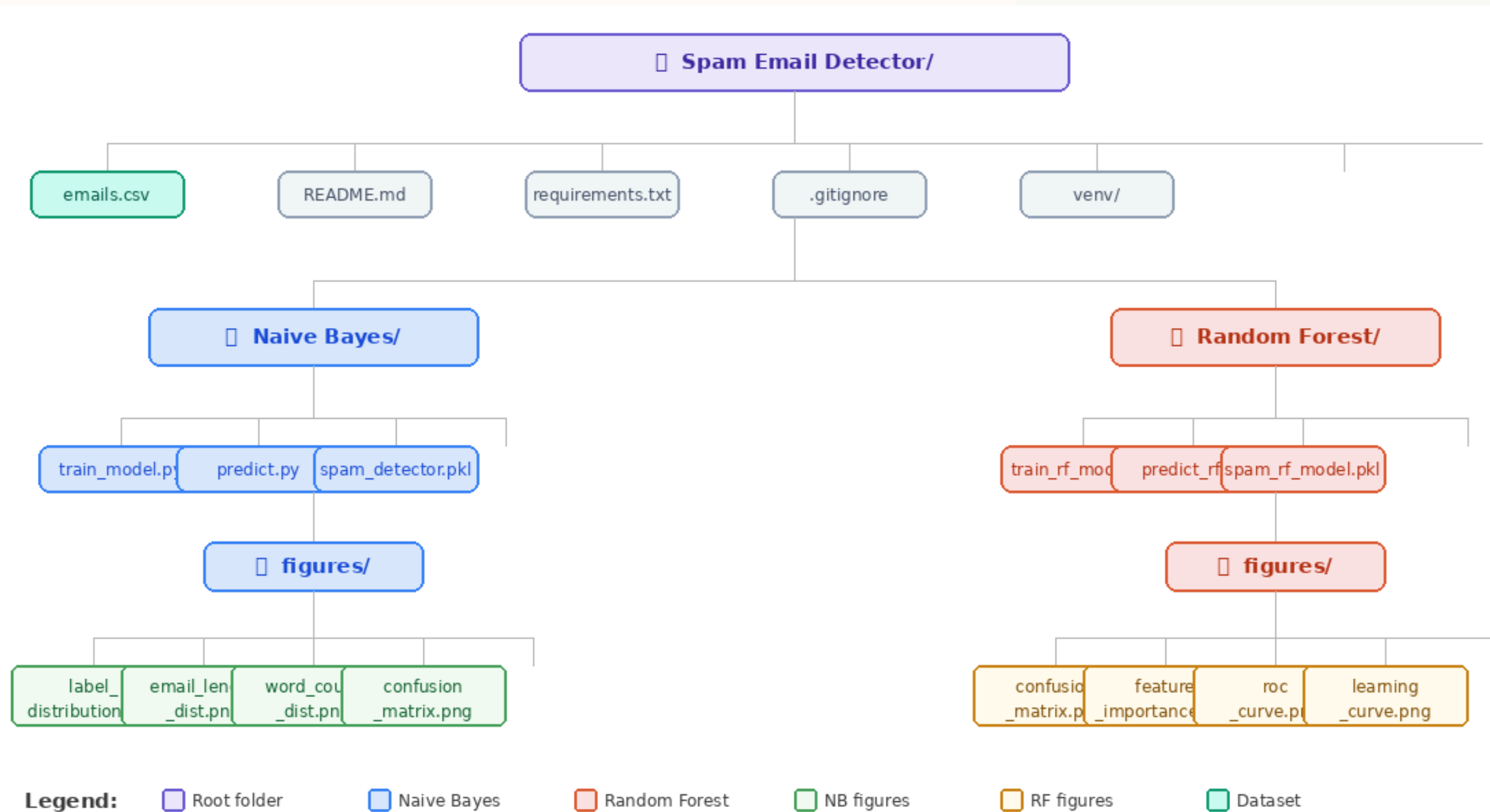
9. Random Forest Classifier

An ensemble learning method that builds multiple decision trees (100 in this project) and merges them for a more stable prediction.

- 🌲 Prevents overfitting via Bagging.
- 🌲 Provides feature importance metrics.
- 🌲 Handles non-linear relationships.
- 🌲 **Accuracy: 98.34%**



10. Directory Structure



Spam Email Detector — Project Directory Structure

11. Output and Results

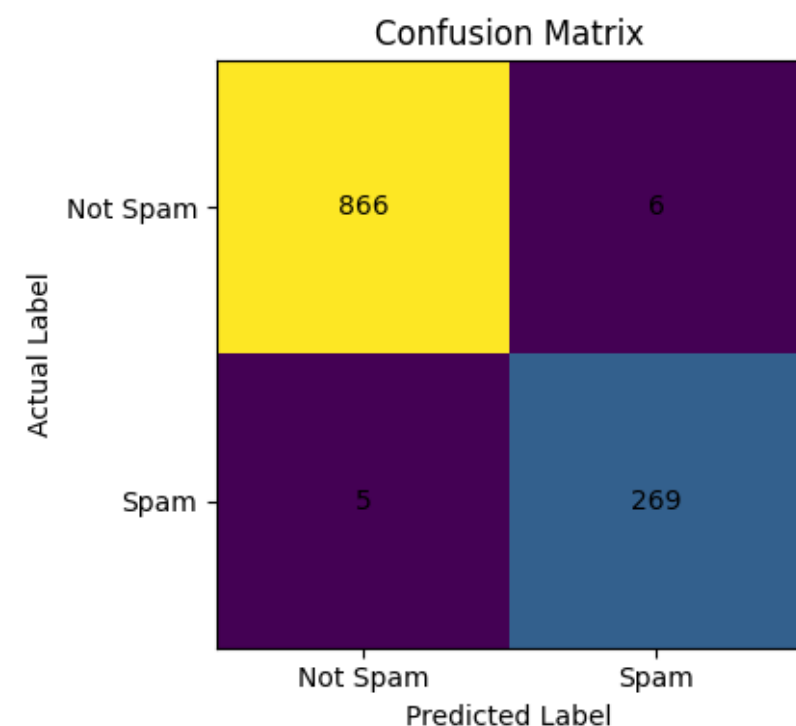
Naive Bayes Classification Report:

```
===== Model Evaluation =====
Model Used: Multinomial Naive Bayes
Accuracy: 0.9904013961605584

Classification Report:
              precision    recall  f1-score   support

   Not Spam      0.99      0.99      0.99        872
        Spam      0.98      0.98      0.98        274

 accuracy              0.99        1146
 macro avg              0.99        1146
weighted avg              0.99        1146
```



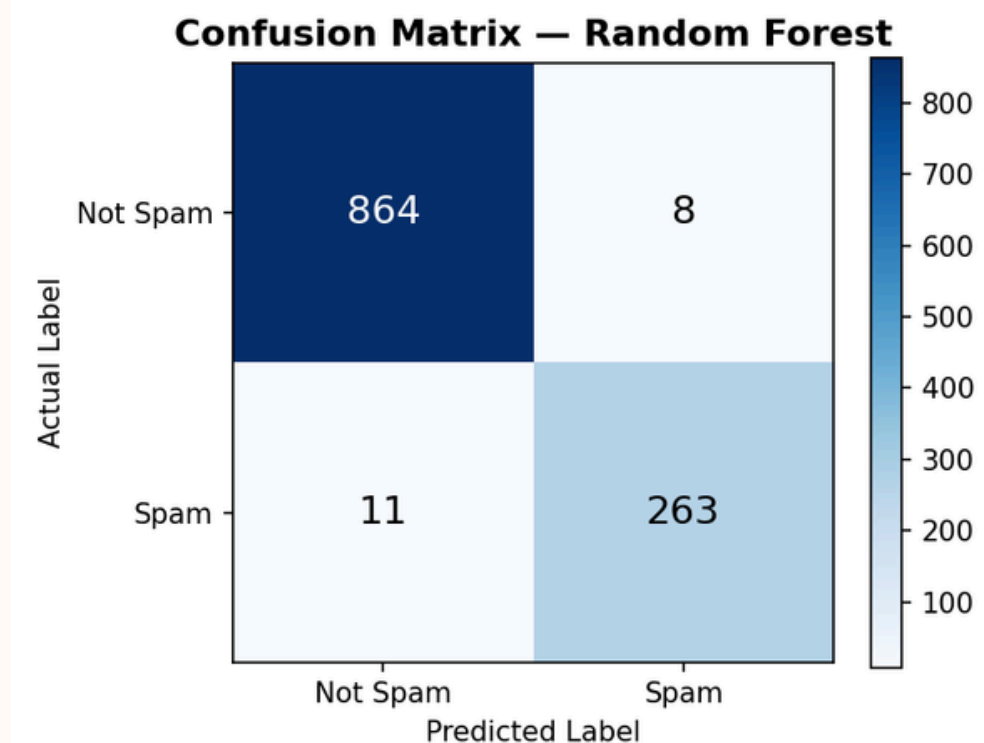
Random Forest Classification Report:

```
===== Model Evaluation =====
Model Used : Random Forest Classifier
Accuracy   : 0.9834205933682374

Classification Report:
              precision    recall  f1-score   support

   Not Spam      0.99      0.99      0.99        872
        Spam      0.97      0.96      0.97        274

 accuracy              0.98        1146
 macro avg              0.98        1146
weighted avg              0.98        1146
```



12. Output and Results(cont.)

Input Email Text	NB Prediction	RF Prediction
Congratulations! You have won a free iPhone. Click the link now to claim your prize.	Spam	Spam
Sir, I have completed the assignment and submitted it before the deadline.	Not Spam	Not Spam
Urgent! Your account has been selected for a \$1000 cash reward. Click here.	Spam	Spam
Dear student, your class test will be held tomorrow at 10 AM.	Not Spam	Not Spam
Your meeting with the supervisor has been scheduled for Sunday at 2 PM.	Not Spam	Not Spam

13. Classifier Comparison

Table 6.1: Performance Comparison: Naive Bayes vs Random Forest

Metric	Naive Bayes	Random Forest
Accuracy	99.04%	98.34%
Not Spam Precision	0.99	0.99
Not Spam Recall	0.99	0.99
Not Spam F1-score	0.99	0.99
Spam Precision	0.98	0.97
Spam Recall	0.98	0.96
Spam F1-score	0.98	0.97
Total Misclassified	11	19
AUC Score	—	0.9983
Training Time	Fast	Slower
Feature Interpretability	Low	High

Situation	Naive Bayes	Random Forest
Large dataset, fast training needed	✓	
Need highest accuracy on this data	✓	
Need feature interpretability		✓
Need AUC / probability calibration		✓
Noisy or complex feature space		✓
Real-time email filtering	✓	

14. Classifier Comparison(cont.)

Accuracy Comparison Graph



Key Metrics Table

Metric	NB	RF
Accuracy	99.04%	98.34%
Misclassified	11	19
Training Time	Fast	Slower
Recall (Spam)	0.98	0.96

15. Conclusion & Future Scope & Limitations

Concluding Remarks

- Both models performed exceptionally well (>98%)
- However, Naive Bayes emerged as the superior model for this specific dataset with 99.04% accuracy and faster execution

Future Improvements

- Use of Deep Learning (BERT/LSTM).
- Web-based UI for user interaction.
- Analysis of attachments and URLs.

Limitations

Only text emails were analyzed.
No attachment or URL analysis.
CLI-only implementation.

Dhonnobad!

Thank you for your
attention.

Any Questions?

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